

Compiler Design

CSE - 420

Assignment 02

Summer 26

Note that:

1. *You will have to submit the soft copy*
2. *Make an cover page which contains you name, student ID and section*
3. *You do not have to submit the ungraded portion.*
4. *Submission form link: <https://forms.gle/ZiPpku3vWN2vX2L26>*

Deadline: 20 July,2026(11:59PM)

Question 1

Consider the following grammar:

1. $D \rightarrow T \text{ id } ; D$
2. $D \rightarrow \epsilon$
3. $T \rightarrow B C$
4. $B \rightarrow \text{int}$
5. $B \rightarrow \text{float}$
6. $C \rightarrow [\text{num}] C$
7. $C \rightarrow \epsilon$

SLR(1) Parse Table:

State	Action								Go - To			
	id	;	int	float	[	]	num	\$	D	T	B	C
0			S4	S5				R2	1	2	3	
1								Accept				
2	S6											
3	R7				S8							7
4	R4				R4							
5	R5				R5							
6		S9										
7	R3											
8							S10					
9			S4	S5				R2	11	2	3	
10						S12						
11								R1				
12	R7				S8							13
13	R6											

Check whether *int [5]a;* is a valid input string according to the grammar.

Ungraded

Show the CLR(1) Parsing table for the above CFG.